

MUHAMMAD SHAHAL

AI/ML Engineer · Full-Stack GenAI & LLM Systems

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PROFESSIONAL SUMMARY

Full-stack AI/ML Engineer with production experience building and deploying scalable AI applications end-to-end. Skilled across **Generative AI (GenAI)**, **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)**, **agentic AI**, and **fine-tuning (PEFT, LoRA, QLoRA)**, with hands-on backend development (**Python, SQL, FastAPI, Flask, Databases, Vector databases**), frontend (**React**), and **cloud-native deployment (AWS, Docker, CI/CD)**. Proficient in **LangChain and LangGraph**, with additional depth in classical ML, statistical modeling, and model evaluation for reliable, production-grade systems.

PROFESSIONAL EXPERIENCE

Sustainability Economics.ai — Generative AI Engineer

2026

- **Grid Tariff Data Automation Platform:** Built an LLM-driven data automation platform to extract and standardize electricity tariff data from utility PDFs and websites across countries, using **AWS S3, DynamoDB, Pydantic schema validation**, feature engineering, country-specific mapping, and unit conversion; reduced manual data-processing effort by **~80%**.
- **Domain-Specific LLM Fine-Tuning & Structured Extraction:** Performed statistical evaluation and model validation on extraction outputs and fine-tuned a **Gemma 3 4B** model using **Unsloth and PEFT** on domain-specific tariff data, improving structured extraction accuracy and enabling more efficient, cost-effective production inference.

Cybpres Innovative Solutions — AI Developer & Data Scientist

2025-2026

- **Agentic AI Code Review System:** Designed the core multi-agent architecture for an LLM-based code review system using **Langchain and LangGraph**, defining a 4-node workflow, code summarization, security/bug detection, quality analysis, and improvement suggestions — contributing foundational architecture to a collaborative agentic AI platform.
- **Real Estate Hybrid Property Matching System:** Engineered **Retrieval-Augmented Generation (RAG)** and hybrid search pipelines on **OpenSearch/FAISS** (BM25 + vector retrieval with reranking); performed model evaluation and containerized the inference service with **Docker**, optimizing for high-performance, low-latency serving.

Luminar Technolab — Data Science Intern

2024-2025

- **ML Pipeline Automation with CI/CD:** Built end-to-end machine learning pipelines with automated **CI/CD via GitHub Actions**, supporting model training, model evaluation, and deployment workflows.
- **NLP Semantic Similarity & Reporting Dashboard:** Developed **NLP** semantic-similarity models for classification and retrieval, applying statistical analysis to validate a **~25%** accuracy gain; delivered **Power BI/Tableau** dashboards that cut manual reporting effort by **~50%**.

CORE SKILLS

Languages, Backend & Frontend: Python, SQL (PostgreSQL/MySQL), React, Bash, FastAPI, Flask, RESTful APIs, gRPC, Microservices, Asynchronous Programming, AI System Design

Generative AI & LLMs: LangChain, LangGraph, RAG, Agentic AI, Prompt Engineering, OpenAI SDK, LLaMA, Vector Databases (Pinecone, FAISS, ChromaDB, OpenSearch), Embeddings, Semantic & Hybrid Search

LLM Fine-Tuning & Optimization: PEFT, LoRA, QLoRA, Unsloth, Fine-Tuning, Quantization, Inference Optimization, vLLM, TensorRT

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, Keras, Model Training, Model Evaluation, Hyperparameter Tuning, Feature Engineering, Statistical Modeling, A/B Testing

Data Engineering & Databases: ETL, Pandas, NumPy, PostgreSQL, MySQL, DynamoDB, MongoDB, RabbitMQ, Redis, Data Pipelines, Data Cleaning

Cloud, MLOps & DevOps: AWS, Azure, Google Cloud (Vertex AI), Docker, CI/CD (GitHub Actions), MLflow, DVC, Airflow, Git

Visualization & Reporting: Power BI, Tableau, Matplotlib, Seaborn, Data Visualization

KEY PROJECTS

Multi-Agent Legal Reasoning System (LLM + RAG + Persistent Memory) ↗

- Designed and implemented a **BNS-aligned** multi-agent **LangGraph** reasoning pipeline with **PostgreSQL**-backed persistent memory and a global immutable retrieval index to prevent statutory mixing and reduce hallucination.
- Built an async **FastAPI** backend with token streaming and thread-isolated **ChromaDB** vector stores, applying model evaluation methodology to validate output reliability.

AI RAG MedBot — Healthcare Knowledge Assistant ↗

- Built a **RAG**-based medical chatbot (**LangChain**, **Groq LLaMA-3-70B**, **Pinecone**, **Hugging Face embeddings**) delivering context-grounded Q&A over hospital doctor-availability data and medical reference documents.
- Architected the pipeline to be **hospital-agnostic** via a swappable **CSV/dataframe** data source, enabling redeployment to new environments with zero code changes.

Resume Matcher Pro — Full-Stack Parsing & Assessment System ↗

- Built a full-stack **resume parsing system** with **FastAPI**, **React**, and **PostgreSQL** for structured PDF/DOCX data extraction and model-backed candidate scoring.
- Implemented **NLP-based** resume-job matching using **spaCy**, **Regex**, and custom scoring algorithms to generate real-time candidate-job match scores.

House Price Prediction — Regression Modeling & Deployment ↗

- Trained and compared **Linear**, **Ridge**, **Random Forest**, **XGBoost**, and **LightGBM** regression models on **180,000+** property listings across 21 Indian cities; selected Random Forest (R^2 **0.87**) after hyperparameter tuning and regularization to reduce overfitting.
- Performed data cleaning, outlier detection (IQR-based filtering), feature engineering, and log-transformation of skewed targets; deployed the final model as an interactive **Streamlit** web app using **Joblib** model persistence.

Movie Recommendation System — Content-Based Filtering with NLP ↗

- Built a content-based recommendation engine using **NLP feature engineering** (tokenization, stemming via NLTK's PorterStemmer) and **CountVectorizer (Bag-of-Words)** over the TMDb dataset, computing cosine similarity across a high-dimensional sparse vector space.
- Serialized precomputed similarity matrices for instant inference and deployed via **Streamlit** for low-latency response times.

EDUCATION

B.Tech, Artificial Intelligence and Machine Learning

Hindusthan College of Engineering and Technology | 2020-2024

CERTIFICATIONS

- IBM Industry Certification Program
- Data Science (NACTET)
- Learnathon ICT

LANGUAGES

English, Malayalam, Hindi